Application of Machine Learning in Predictive Coronary Heart Disease (MICHD) Diagnosis

Paul Barbereau, Amandine Dejardin, Samuel Perry
School of Computer and Communication Sciences, EPFL, Lausanne, Switzerland

***Abstract*—Cardiovascular diseases (CVDs) are a leading cause of mortality worldwide, making early risk screening from clinical and lifestyle indicators a key public health goal. We present a machine learning pipeline to predict myocardial infarction or coronary heart disease (MICHD) from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset (328,135 training and 109,379 test individuals across 321 variables). To address missing data, survey-specific coding, feature redundancy, and severe class imbalance (8.83% positive rate), we combine domain-informed selection of 47 clinically relevant features with codebook-guided recoding, fold-wise median imputation, standardization, and ℓ_2-regularized logistic regression with threshold tuning. In 5-fold cross-validation, our model achieves an F1 score of 0.4199 ± 0.0030 and $87.15\% \pm 0.05\%$ accuracy, substantially outperforming uncleaned baselines and linear regression models.**

I. Introduction

Early identification of individuals at risk of myocardial infarction or coronary heart disease (MICHD) using non-invasive health surveys can enable timely preventive care. In this work, we predict binary MICHD diagnosis ($y \in \{-1, 1\}$) from the CDC's 2015 BRFSS telephone survey [2]. Applying machine learning directly to the raw dataset presents four main challenges: (1) **high dimensionality** ($D = 321$ columns, including administrative metadata unrelated to health); (2) **survey-specific sentinel codes**, where "None/Zero," "Don't know," and "Refused" are encoded as large integers ($7, 9, 77, 88, 99, 888, 9999$) that distort numerical distributions; (3) **multicollinearity** among raw and derived variables; and (4) **class imbalance**, as only 28,975 of 328,135 training respondents (8.83%) report MICHD. To address these issues without external libraries, we implement six foundational algorithms in `implementations.py` and pair ℓ_2-regularized logistic regression with codebook-driven preprocessing and 5-fold cross-validation in `run.py`.

II. Models and Methodology

A. Model Selection and Baselines

We implemented the six core methods in `implementations.py`: linear models minimizing mean squared error (`least_squares`, `ridge_regression`, `mean_squared_error_gd`, and mini-batch $B = 1$ `mean_squared_error_sgd`) and probabilistic classifiers minimizing negative log-likelihood (`logistic_regression` and `reg_logistic_regression`). Because MICHD diagnosis is binary, we selected ℓ_2-regularized logistic regression (with $\lambda = 0$ corresponding to standard logistic regression). Given a feature vector $\mathbf{x}_i \in \mathbb{R}^{D^*}$ and sigmoid $\sigma(z) = (1 + e^{-z})^{-1}$, the positive class probability is $p(y_i = 1 \mid \mathbf{x}_i) = \sigma(\mathbf{w}^T \mathbf{x}_i)$, and the regularized training loss is:

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i \log \sigma(\mathbf{w}^T \mathbf{x}_i) + (1 - y_i) \log \big(1 - \sigma(\mathbf{w}^T \mathbf{x}_i)\big) \Big] + \lambda \|\mathbf{w}\|_2^2, \tag{1}$$

where $\lambda \geq 0$ penalizes large weights to prevent overfitting.

B. Data Preprocessing

Our preprocessing pipeline in `run.py` transforms the 321 raw columns into a clean design matrix of $D^* = 48$ features (47 variables plus an intercept):

1) **Feature Selection:** Using the BRFSS codebook [1], [2], we selected 47 variables (`cvd_variables`) covering general health (`GENHLTH`, `PHYSHLTH`), cardiovascular history (`BPHIGH4`, `TOLDHI2`, `CVDSTRK3`, `HAREHAB1`), diabetes (`DIABETE3`), demographics (`SEX`, `_AGE80`, `EDUCA`, `INCOME2`), and lifestyle factors (smoking, alcohol, exercise, BMI).
2) **Survey Code Recoding:** In `preprocess_cols`, codes representing "None" or zero days/drinks $(88, 888)$ are mapped to 0, while "Don't know," "Refused," or missing codes $(7, 9, 77, 98, 99, 777, 999, 7777, 9999, 3, 14)$ are set to `NaN`.
3) **Imputation and Standardization:** In `preprocessing_X`, all-`NaN` and constant columns are removed, and remaining `NaN` values are imputed with the training column median. Features are standardized using the training mean and standard deviation to ensure stable gradient descent, a bias column of ones is added, and labels $y \in \{-1, 1\}$ are mapped to $\{0, 1\}$.

C. Optimization, Hyperparameter Tuning, and Prediction

Weights $\mathbf{w}$ are optimized over 1,000 gradient descent iterations via $\mathbf{w} \leftarrow \mathbf{w} - \gamma \nabla \mathcal{L}(\mathbf{w})$, where:

$$\nabla \mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^{N} \big(\sigma(\mathbf{w}^T \mathbf{x}_i) - y_i\big) \mathbf{x}_i + 2\lambda \mathbf{w}. \tag{2}$$

Test predictions are obtained by thresholding the predicted probability at τ: $\hat{y}_i = 1$ if $\sigma(\mathbf{w}^T \mathbf{x}_i) \geq \tau$ and -1 (or

0) otherwise. We tuned (λ, γ, τ) via 5-fold cross-validation (`SEED = 67`) maximizing the F1 score:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{3}$$

with Precision $= \frac{\text{TP}}{\text{TP+FP}}$ and Recall $= \frac{\text{TP}}{\text{TP+FN}}$. Imputation and standardization were computed inside each fold to prevent data leakage. Grid search over $\lambda \in \{0\} \cup [10^{-6}, 10^{0}]$, $\gamma \in [10^{-3}, 10^{0}]$, and $\tau \in [0.1, 0.7]$ yielded $\lambda = 10^{-6}$, $\gamma = 0.775$, and $\tau = 0.20$.

III. Results

A. Comparison with Baseline Models

Table I compares all six implemented algorithms on the cleaned 47-feature dataset across 5 cross-validation folds. Unregularized `least_squares` fails with a singular matrix because derived BRFSS variables (e.g., `_AGEG5YR`, `_AGE65YR`, `_AGE80`, `_AGE_G`) are collinear, making $\mathbf{X}^T\mathbf{X}$ rank-deficient. Adding an ℓ_2 penalty in `ridge_regression` ($\lambda = 10^{-6}$) restores invertibility and matches full-batch `mse_gd` (F1 $= 0.4109$ at $\tau = 0.20$). Meanwhile, `mse_sgd` ($B = 1$, 1,000 steps) observes only 0.38% of each fold and suffers from high variance (F1 $= 0.3106 \pm 0.0852$). Both `logistic_regression` and `reg_logistic_regression` at $\tau = 0.20$ outperform all linear MSE baselines (F1 $= 0.4199$ vs. 0.4109, accuracy 87.15% vs. 85.76%), confirming that the log-loss objective and bounded sigmoid output better capture binary disease risk.

TABLE I
5-Fold CV Comparison on Cleaned 47 Features (Mean ± SD)

Method	τ	F1 Score	Accuracy	Prec.	Rec.
`least_squares`	–	*Singular matrix ($\sigma_{\min} = 2.6 \times 10^{-10}$)*			
`ridge_reg` (λ=10^{-6})	0.50	0.0766 ± 0.0021	0.9140 ± 0.0011	0.734	0.040
`ridge_reg` (λ=10^{-6})	0.20	0.4109 ± 0.0032	0.8576 ± 0.0010	0.324	0.563
`mse_gd` (γ=0.1)	0.50	0.0767 ± 0.0021	0.9140 ± 0.0011	0.733	0.041
`mse_gd` (γ=0.1)	0.20	0.4109 ± 0.0030	0.8576 ± 0.0010	0.324	0.562
`mse_sgd` (γ=10^{-3})	0.20	0.3106 ± 0.0852	0.8836 ± 0.0196	0.367	0.328
`logistic_reg`	0.50	0.2185 ± 0.0098	$\mathbf{0.9146 \pm 0.0012}$	0.568	0.135
`logistic_reg`	0.20	$\mathbf{0.4199 \pm 0.0030}$	0.8715 ± 0.0005	0.349	0.527
`reg_log_reg` (λ=10^{-6})	0.50	0.2185 ± 0.0098	$\mathbf{0.9146 \pm 0.0012}$	0.568	0.135
`reg_log_reg` (λ=10^{-6})	**0.20**	$\mathbf{0.4199 \pm 0.0030}$	0.8715 ± 0.0005	0.349	0.527

B. Ablation Study

Table II isolates the impact of each pipeline modification. Compared to using all 315 non-constant features (Config. 1), restricting to our 47 domain features (Config. 2) cuts dimensionality by 85% and training time by $6.5\times$. Adding BRFSS survey code cleaning (Config. 3) lifts F1 at $\tau = 0.50$ from 0.1765 to 0.2185 ($+4.20$ points). Finally, tuning the threshold from $\tau = 0.50$ to $\tau = 0.20$ (Config. 6) provides a $+20.14$ point F1 jump ($0.2185 \rightarrow 0.4199$), outperforming the 315-feature model at $\tau = 0.20$ (Config. 5, 0.4149) in both F1 and accuracy.

IV. Discussion

Class Imbalance and Thresholding: Because only 8.83% of patients have MICHD, predicting -1 everywhere yields 91.17% accuracy but F1 $= 0$. At $\tau = 0.50$, logistic regression achieves 91.46% accuracy but only 13.53% recall (F1 $= 0.2185$). Lowering τ to 0.20 accounts for the low positive prior, nearly quadrupling recall to 52.69% and doubling F1 to 0.4199. Clinically, higher recall is critical in screening, where missed heart disease (false negatives) is far costlier than follow-up testing (false positives).

TABLE II
Ablation Study of Pipeline Modifications (5-Fold CV)

Pipeline Configuration	D^*	CV F1 Score	CV Accuracy
1. All raw features, no code clean, $\tau = 0.50$	316	0.1814 ± 0.0067	0.9142 ± 0.0007
2. 47 features, no code clean, $\tau = 0.50$	48	0.1765 ± 0.0068	0.9135 ± 0.0008
3. 47 features + code clean, $\tau = 0.50$	48	0.2185 ± 0.0098	$\mathbf{0.9146 \pm 0.0012}$
4. 47 features, no code clean, $\tau = 0.20$	48	0.4113 ± 0.0039	0.8671 ± 0.0009
5. All raw features, no code clean, $\tau = 0.20$	316	0.4149 ± 0.0020	0.8659 ± 0.0005
6. Full Pipeline (47 + clean, $\tau = 0.20$)	**48**	$\mathbf{0.4199 \pm 0.0030}$	0.8715 ± 0.0005

Clinical Interpretability and Limitations: On the standardized training set (final log-loss 0.2231, intercept $w_0 = -3.254$), the strongest predictors of MICHD are poor general health (`GENHLTH`, $w = +0.455$), older age (`_AGE80`, $w = +0.442$; `_AGEG5YR`, $w = +0.405$), hypertension (`_RFHYPE5`, $w = +0.142$), high cholesterol (`_RFCHOL`, $w = +0.137$), male sex (`SEX`, $w = -0.359$), prior heart rehabilitation (`HAREHAB1`, $w = -0.262$), and stroke (`CVDSTRK3`, $w = -0.184$), matching established medical risk factors. Limitations include standardizing categorical variables without one-hot encoding, retaining collinear age features, and assuming a linear boundary. Future work could add one-hot encoding, ℓ_1 feature selection, polynomial interactions, and class-weighted loss.

References

[1] Duke University, "BRFSS 2015 Project Codebook," STA 102 Course Materials, https://www2.stat.duke.edu/~cr173/Sta102_Sp16/Proj/codebook.html.

[2] CDC, "Behavioral Risk Factor Surveillance System 2015 Codebook Report," 2016.